

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular End Semester Examination – Winter 2022

Course: B. Tech. Branch : Civil Engineering Semester : 7th Sem

Subject Code & Name: BTCVC 702_ Infrastructure Engineering

Max Marks: 60

Date: 30.01.2023

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/ Marks
CO)

Q. 1 Solve any two of the following.

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|--|-------------|----------|
| A) Define permanent way and explain the components of a permanent way with sketch. | CO 1 | 6 |
| B) Define rail. State its functions and Explain types of rails with suitable diagram. | CO 1 | 6 |
| C) Define Airport and Airport Engineering. Explain components of airport. | CO 2 | 6 |

Q.2 Solve any two of the following.

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|--|-------------|----------|
| A) Define the following terms:
a) Coning of wheel
b) Adzing of sleeper
c) Tilting of Rails
d) Station
e) Gauge
f) Ballast | CO 1 | 6 |
| B) Draw layout for simple turnout and explain any four components of Turnout. | CO 4 | 6 |
| C) Explain types of railways crossing with neat labelled diagram. | CO 1 | 6 |

Q. 3 Solve any two of the following.

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|---|-------------|----------|
| A) Describe the full-face method of Tunneling in hard rock with neat labelled diagram. | CO 3 | 6 |
| B) Sketch the various types of tunnels cross-section. State under what conditions each is adopted. | CO 3 | 6 |

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|---|-------------|----------|
| C) Define the following terms related with tunnel Engineering | CO 1 | 6 |
| a) Crown b) Drift c) Invert | | |
| d) Heading e) Portal f) Spring line | | |

Q.4 Solve any two of the following.

- | | | |
|---|-------------|----------|
| A) Define : a) Docks b) Harbour .State factors affecting on site selection for docks and harbour. | CO 1 | 6 |
| B) Draw a sketch showing components of harbour. Define any two of them. | CO 3 | 6 |
| C) Explain any six components of bridge. | CO 2 | 6 |

Q. 5 Solve any two of the following.

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|--|-------------|----------|
| A) Define tide. Explain types of tide and state uses of tides. | CO 2 | 6 |
| B) State requirements of good harbour and explain classification of harbour based on utility and location. | CO 1 | 6 |
| C) What is Airport and explain the characteristics of Aircraft and Explain types of runways with sketches. | CO 2 | 6 |

***** End *****